

Photoelectric Effect Lab

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PHY4822L, Fall 2026

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September 15, 2026

Abstract

The discovery of the photoelectric effect was essential in ushering in the quantum paradigm shift in physics. Using this effect we describe Planck's constant (h) and the work function (ϕ). By identifying the stopping voltage (V) where photocurrent is measured at zero we can calculate a value for h and ϕ . Photocurrent in this experiment is generated by ultraviolet photons produced by a mercury bulb incident on a phototube. The experiment also uses filters to isolate particular wavelengths of light. h and ϕ are determined by fitting a line to a graph of V vs λ^{-1} (or inverse of the wavelengths of light). The value acquired for h was $(6.58 \pm 4.8) \times 10^{-34} \text{ J} \cdot \text{s}$, for ϕ , $1.58 \pm 1.84 \text{ eV}$. This agrees with the accepted value of h $6.62607015 \times 10^{-34} \text{ J} \cdot \text{s}$ [7]. The value of ϕ is consistent with cesium-antimony (Cs-Sb) photocathodes [3]. We evaluated our uncertainty with both a standard and modified bootstrap method. The large uncertainty evaluated by this method will be explained as well as how our experimental techniques could have been optimized.

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1 Introduction

It is only through experiment that we can understand how a photon, a massless packet of energy, may interact with electrons that are traditionally unobservable point particles. The discovery of the photoelectric effect would win Albert Einstein the Nobel Prize for physics in 1921 [4] however this effect was observed much earlier by Heinrich Hertz. Hertz understood that metal was conductive, therefore a burst of energy may knock loose electrons off of the metal. In 1900, Max Planck discovered that electron's energy can be modeled discretely. The electrons in Planck's experiment only emit or absorb energy in quanta: $E_{\text{quanta}} = nh\nu$, where n is an integer and h is a new fundamental constant [7]. Einstein would use this relation to justify how photons with sufficient energy may produce the photoelectric effect. While Planck viewed using h as just a mathematical trick in order to explain his experimental results, it would later be understood to be the minimum allowed action or physical change in the universe. The current accepted value for h is $6.62607015 \times 10^{-34} \text{ J}\cdot\text{s}$.

2 Background

2.1 Determining Planck's Constant

To find the value of Planck's Constant we must start with the energy of the photons from the mercury bulb. The photons incident on the phototube have a quantized energy given by

$$E_p = h\nu \quad (1)$$

where E is the energy of single photon, h is Planck's Constant, and ν is the frequency of the mercury bulb's light. Photons with enough energy will liberate electrons from the photocathode. These electrons will have energy

$$E_e = E_p - \phi = h\nu - \phi \quad (2)$$

ϕ is the work function of the metal which contained the electrons. The work function is the minimum amount of energy required to remove an electron from the aforementioned metal. For our experiment we want to measure the maximum kinetic energy as we are applying a reversing voltage.

$$E_e = K_{\text{max}} = eV_{sp} \quad (3)$$

Here e is the charge of an electron (1.602×10^{-19} coulombs) and V_p is potential when the photocurrent goes to zero. Substituting this into equation (2), and recalling $\nu = \frac{c}{\lambda}$ where c is the speed of light ($2.998 \times 10^8 \text{ m/s}$) and λ is the wavelength of light.

$$eV_{sp} = h\nu - \phi = \frac{hc}{\lambda} - \phi \quad (4)$$

rearranging for h

$$h = \frac{\lambda}{c}(eV_{sp} + \phi) \quad (5)$$

In our experiment we will graph V_s vs $\frac{1}{\lambda}$ in order to determine this value, thus we state

$$h = \frac{\text{slope} \times e}{c} \quad (6)$$

2.2 Inverse Square Law

We also must understand the relationship between the distance of our light source and its relation to the photocurrent we measure. We know our bulb will radiate photons in every direction giving us:

$$\text{Intensity} = \frac{\text{Power}}{4\pi r^2} \quad (7)$$

r is the distance of the bulb to our photocathode. The intensity of the photons is proportionate to the count of photons, we can infer that the photon count is inversely proportional to our distance.

$$n_\gamma \propto \frac{1}{r^2} \quad (8)$$

it follows that the number of electrons knocked loose by the incident photons are also proportional

$$n_e \propto n_\gamma \quad (9)$$

the number of electrons being freed from the metal that causes the change in photocurrent, thus:

$$I \propto \frac{1}{r^2} \quad (10)$$

this is the result we observed later in our data and more generally referred to as the inverse square law.

3 Experimental Techniques



Figure 1: Experimental setup, the EP-07 is in the foreground on the mechanical elevator, the digital ammeter and voltmeter to the right of the detector, and the ES-18 in the background

First, we positioned the ES-18 Mercury Bulb 10 cm away from the front lens of the EP-07 Precision Photoelectric Effect detector. We then powered on the bulb and the power supply for the detector. The detector, while equipped with an analog ammeter, is connected to a digital ammeter and voltmeter. We decided to use the digital counterparts regardless of the noise introduced due to ease of measurement. Finally, before beginning measurements we ensured the cap of the mercury bulb was as even as possible relative to the detector. With all equipment physically placed we ensured that we were applying no retarding voltage to the detector and that the red light emitted by the analog ammeter was obscured from reflective elements on the setup adjacent to the mercury bulb. Finally, we shut off the overhead light.

With no light contaminants, we began our first measurement by measuring the photocurrent at different distances, moving the bulb further away from the detector in increments of 1 cm. We were careful to center the bulb cap after each movement. This was repeated a total of twenty times, moving the bulb 30 cm away from the detector.

Moving the bulb to 25 cm, we then proceeded with our first measurement of the

stopping voltage. Starting from zero, we increased the gate voltage setting provided by the detector in increments of .2 volts until we reached 4 volts, keeping with our standard of twenty measurements.

As mentioned earlier, the digital ammeter/voltmeter are precise but extremely sensitive. In our experiments we found it difficult to set voltage exactly or measure photocurrent with certainty. We decided to observe the variance of photocurrent over our measurements both when adjusting the knob for gate voltage and when reading the dynamic photocurrent measurements. This led us to see a common variance of $\pm 0.5 \mu A$.

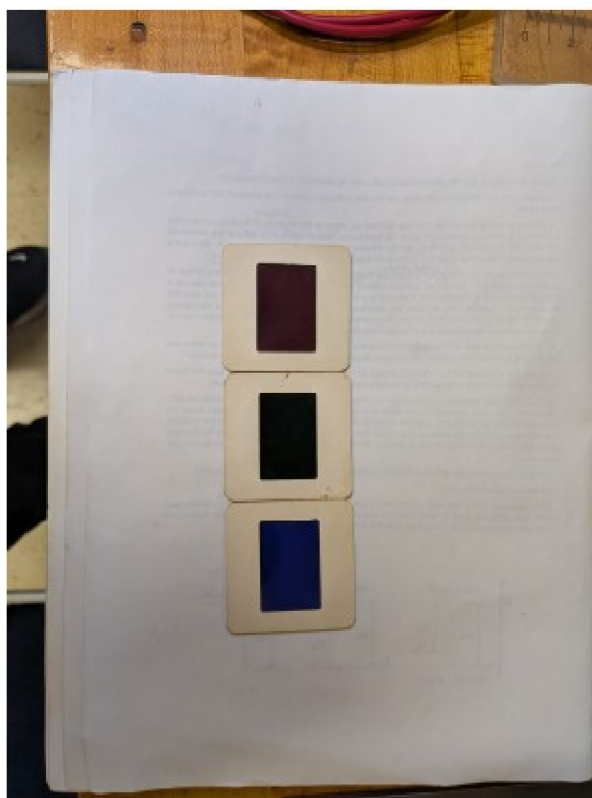


Figure 2: The red green and blue gel filters

Provided by the lab are three different light filters, red, green, and blue. The detector allows us to attach every filter with a metal clamp, and we paid due diligence to how we fit the filters by ensuring that neither the metal clamp nor the white paper edges of the filters obscured the lens on the detector. For every filter, we then repeated the previous procedure used for no filter. We followed this up with a final measurement with the red filter at 50 cm.

4 Data

The wavelength of light from the mercury bulb is 254nm. The wavelengths of light after applying the red, green or blue filter are 590, 546, 405nm respectively. Note, these are not the traditionally accepted wavelengths of light for those specific filter colors, it is the wavelength perceived by the detector. As shown in the graphs below we were careful to ensure the distances below are accurate, however, the distance will not impact our measurement of the stopping voltage. As one may deduce from the background section, the energy of the incident photons does not depend on distance.

The following graphs are in the order in which we took data.

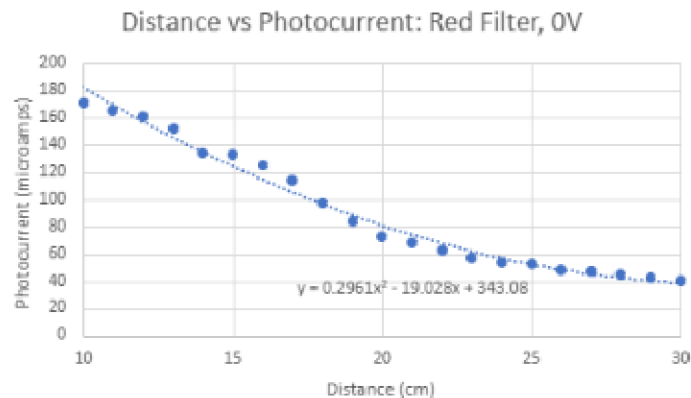


Figure 3: Fitted to this data is a trend line which fails to show the inverse square law, this is due to the issues outlined later about the warming up of the mercury bulb, which prevents the steep drop-off we expect

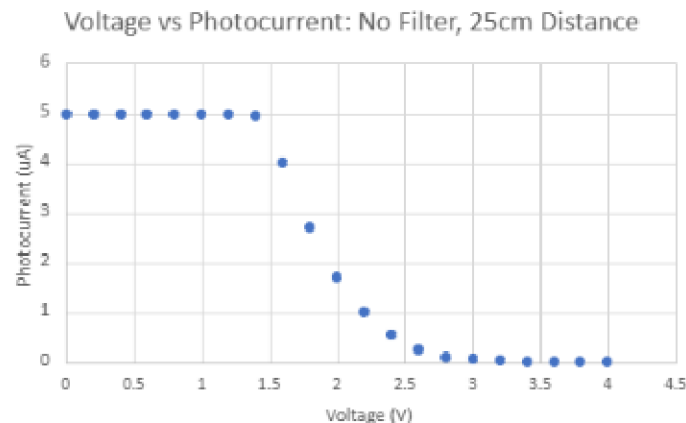


Figure 4: Note how in this image the initial section is almost decreasing linearly, this is not expected and is due to the distance the bulb was placed away from the detector. More will be discussed in later sections. Unlike the other graphs the vertical axis is actually measured in milliamperes not microamperes

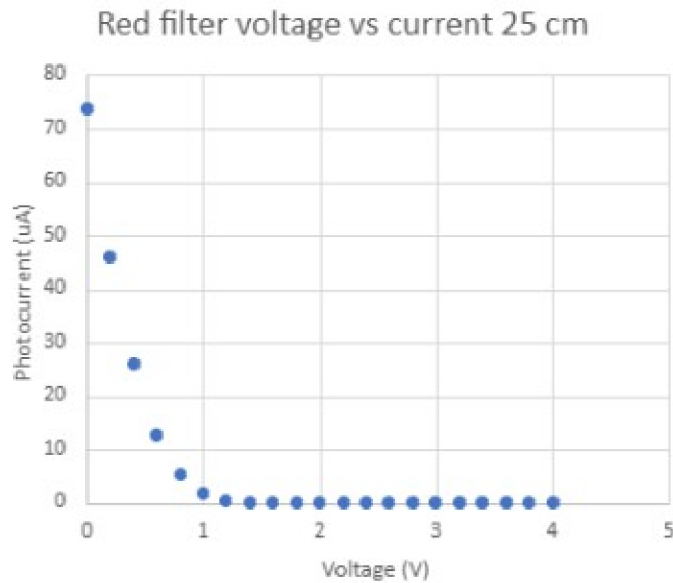


Figure 5: Unlike the last graph we finally see the curve outlined in the detector user manual[2]

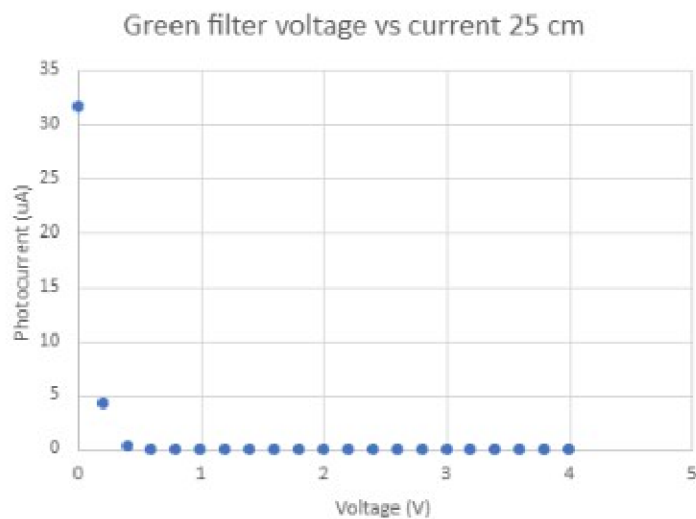


Figure 6: The green filter should technically be the second lowest photocurrent detected as it is the second closest wavelength to the mercury bulb. This could be due to a couple of variations discussed later

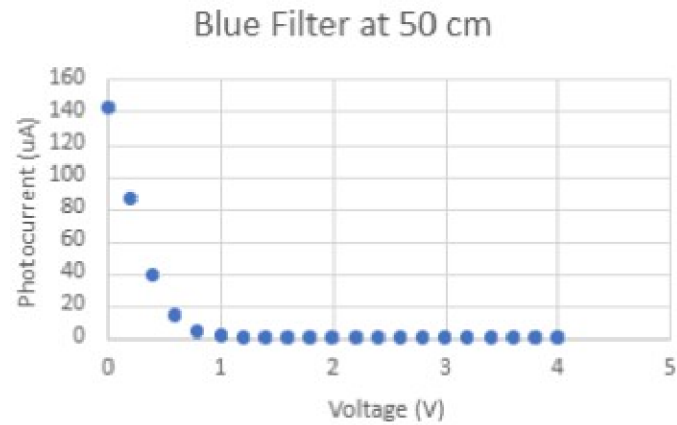


Figure 7: The blue filter, closest in wavelength to the mercury bulb, has the highest measured photocurrent of the three filters

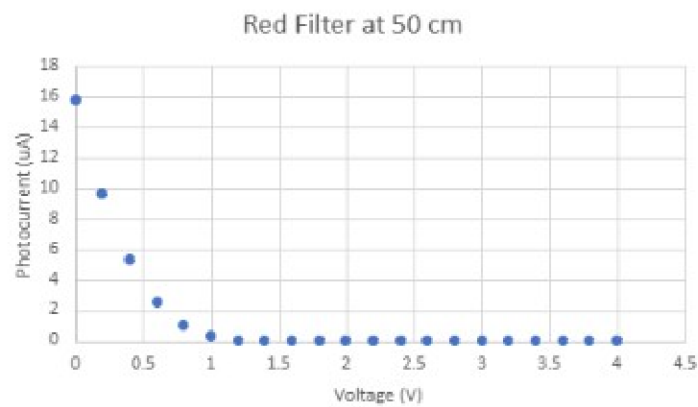


Figure 8: The photocurrent, as expected, starts at roughly a fourth of what the red filter at 25cm starts at. Twice the distance, a fourth the intensity. As our earlier math tells us, the stopping voltage will be the same as the first red filter run

5 Analysis

5.1 Determining h

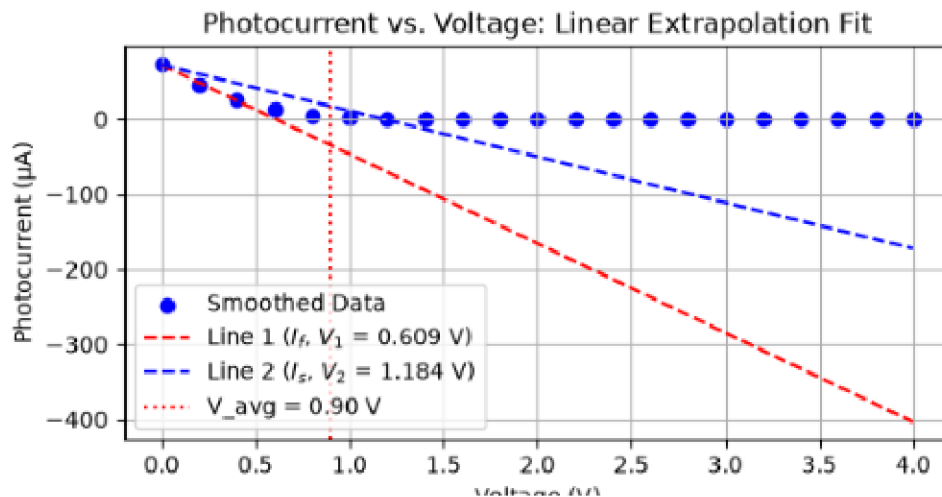


Figure 9: depiction of the two methods to determine V_s and the resulting average

The first step immediately after collecting data was to zero the photocurrent readings. We know for certain that our data contained a valid cutoff voltage and that theoretically our photocurrent should go to zero. Using the digital ammeter, we would observe that consistently after a certain voltage we could no longer decrease the photocurrent. To accomplish this, we input our data into Excel, identified where we measured a consistent reading and called this value zero. This is reflected in the above graphs.

Important to further analysis, these values do not give an explicit intercept with the voltage axis. We cannot say for certain that any given point is the stopping voltage so we must estimate. We decided that we should measure the voltage in two ways; an underestimation and an overestimation. Graphically, we visualized two intersecting lines, one with a high negative slope and another with a smaller negative slope and our stopping potential would be somewhere in between the two horizontal intercepts. It would appear as if someone took a navigation divider and lined the left side up to the near vertical high slope region and let the right side intersect the near horizontal voltage region.

Using the python code in Appendix A we calculated our value of h and ϕ . First, we needed to separate the regions of fast and slow decay. We took the first derivative of the data $\frac{dI}{dV}$ and found the maximum decay value. We set a limit that anything above half of this value defines the high decay region and anything below is the slow decay region. These two regions are almost linear in nature so we decided to perform a linear regression for the high and low regions independently. For the fast decay region we find where the x intercept of the line is, calling this value V_1 . V_1 is purposely an underestimation. Next, we identified the voltage at which the line fit of the high and

low regions intercept, this intercept is the knee of the graph. by adding the voltage value that defines the knee's location to V_1 we find our overestimate value V_2 . By averaging these two values, we wind up with our estimation of the stopping voltage for that particular run.

With our five estimates of stopping voltage, we can now perform a line fit of V vs

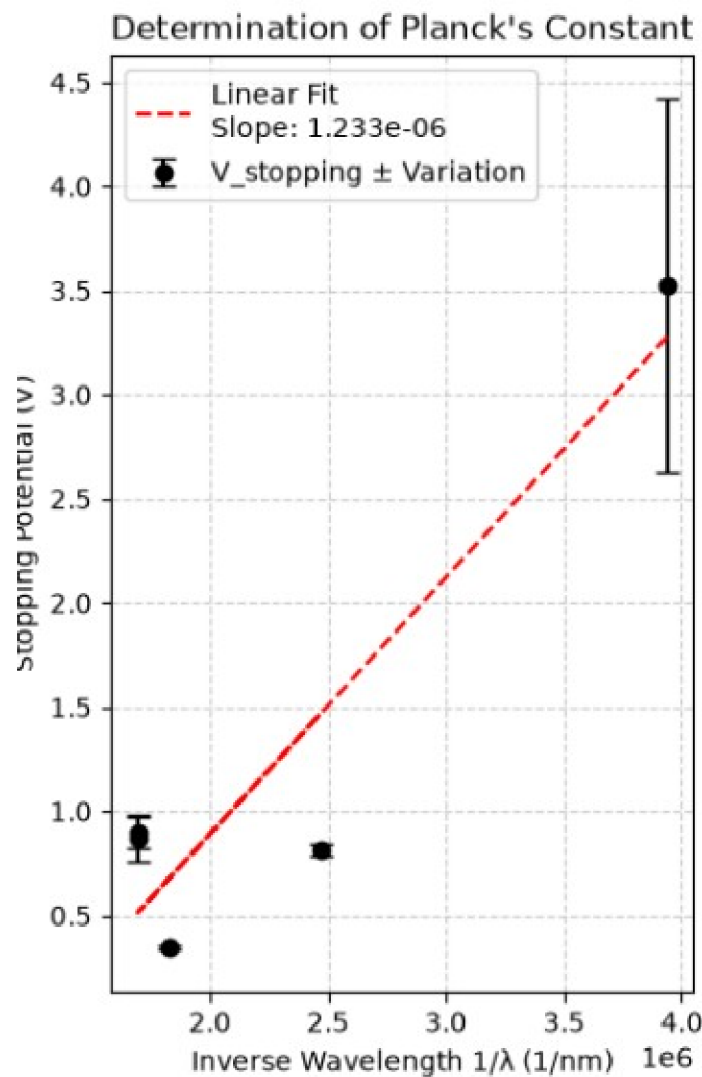


Figure 10: Line fit for the five runs, the slope is then multiplied by electron charge and divided by the speed of light to obtain h

5.2 Determining Uncertainties

As seen in figure 10, there are uncertainties estimated for each parameter. To propagate our uncertainty of the photocurrent to the stopping voltage, we used the bootstrap method[5]. This function is in Appendix A. The idea behind the bootstrap method is to propagate uncertainty using a given deviation in measurement. This can be described as a Monte-Carlo simulation. It operates by varying the photocurrent over the given uncertainty and logging the resultant stopping voltage. We can take the standard deviation of these values which provides us the value of uncertainty for the stopping voltage. We can then use this function again to find the uncertainty for Planck's constant and the work function.

6 Discussion

6.1 Errors in the Experimental Setup

The most obvious result of our data is the large uncertainty in our calculations of h yet a rather small uncertainty for the stopping voltage. To reiterate, h was $(6.58 \pm 4.8) \times 10^{-34} \text{ J} \cdot \text{s}$, for ϕ , $1.58 \pm 1.84 \text{ eV}$. We have the benefit of knowing that our direct calculation of h is within one percent error and the work function is accurate to the expected range for a common material used in photocathodes. Yet the range of our uncertainty, specifically for the work function which states that the work function could possibly be negative, is excessive. The clearest reason is that our code is designed to output results within two standard deviation. This ensures that we are 95% certain that our actual value is within the bounds of our uncertainty.

As seen in figure 10, this trend does not follow for every one of our deviations in stopping voltage. Every data point other than the run without the filter carry incredibly narrow deviations which would imply that, even at two standard deviations, we are almost assured of these readings of stopping voltage. This contradicts the reality of the experiment and highlights some of the mistakes we made when recording our data. In figure 3, we clearly see the slope we expect from the inverse square law but with significant deviation in the first few data points which is contrasted by the second half of the graph where we see a more expected grouping of values for photoelectrons. This is attributed to two things; we failed to give the mercury bulb sufficient time to heat up, and the bulb started too close to the detector. For the first point, the device manual for the EP-07[2] explicitly states we must wait a minimum of thirty minutes for the Hg bulb to stabilize. While we understood that it must be given time, we only offered it about fifteen which inevitably led to shaky measurements and potentially allowed the bulb's intensity to increase linearly with the distance to the detector. For the second point, the detector is incredibly sensitive to even small changes but this would imply that we may accidentally overload the detector. Being this close allows for photons that had been returned to the bulb's surface to again be reflected towards the detector. This leads to the explanation of uncertainty for the run in figure 4, the almost linear quality of the initial readings was a clear sign the detector was essentially being overexcited with photons until the gate voltage was sufficient to return the decay we expected. This naturally caused some incredible deviation for the fast decay region.



Figure 11: While the casing is good at dealing with reflections, it is not perfect

Even with this, we clearly have not taken into account some other uncertainty as the runs conducted with filters seem to be more confident. Unlike the no filter runs, we are ideally removing all UV light by only allowing wavelengths above 254nm to pass. For context, because the mercury in the bulb has an allowed electronic transition where the energy difference is about 4.89 eV, the emitted photon carries exactly that energy. excited mercury also allows for transitions that produce visible light which is why the filters are useful.

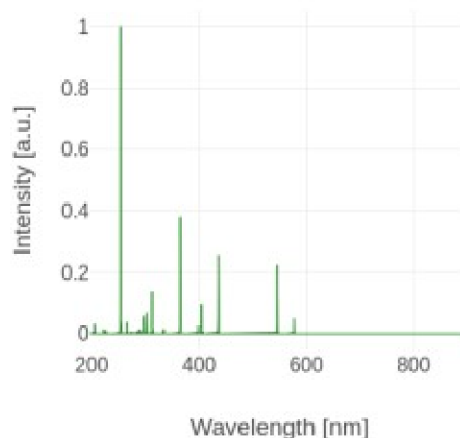


Figure 12: Relative emission lines, shows the clear spike in the UV range[1]

If we assume the filters are perfect then we would expect the photocurrent to match the intensity seen in the above spectra but that isn't the case. The green light is

clearly the least intense of the three runs at 250nm. This indicates that the filters, most noticeably the red filter, are leaking some UV light into the detector.

6.2 Conclusions Drawn from Graphical Analysis

Because we know that photocurrent vs distance in figure 3 is an inverse square, we can linearize the data set by setting the x axis to $1/r^2$ giving us:

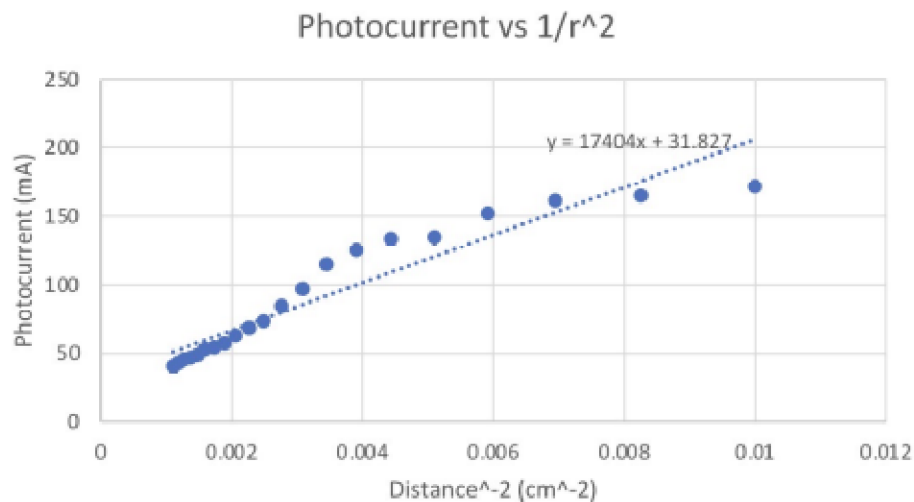


Figure 13: The same data as figure 3, it is still impacted by the issues presented in the previous section

The slope is equal to $17404 \text{ mA} \cdot \text{cm}^2$. This slope corresponds to the sensitivity of the detector. The units are $\text{current} \cdot \text{area}$ which can be made into $\frac{\text{number of electrons}}{\text{time}} \times \text{area}$ which tells us that this is calculating the number of electrons knocked loose over some scanning area.

Explained in the analysis section, we consider two different regions in this data. We must explore why the graph behaves this way. In a perfect detector the anode does not contain electrons, but the real anode has a work function. The bulb is capable of knocking electrons free from the anode too and they will travel back to the cathode. Without this we would have a graceful exponential decay but instead our cutoff voltage is underestimated. Along with this we should expect to see a constant negative slope as the electrons knocked free outnumber the electrons moving in the expected direction. In our data we did not observe this as the noise recorded from the digital ammeter blurred this decrease. In regard to our lab we will be underestimating the value of h because of the stopping voltage even if everything else is conducted perfectly. This is manifested in our results for h .

With this understanding we find that our measured work function corresponds to the anode (ϕ_a) rather than the photocathode (ϕ). Because the photocathode and anode are made of dissimilar materials connected in a circuit, electrons flow through the

external wiring until their Fermi levels equalize [6]. This creates what is called a Contact Potential Difference (CPD):

$$\text{CPD} = - \frac{\phi_a - \phi_c}{e} \quad (11)$$

where ϕ and ϕ_c are the work functions of the anode and cathode, respectively. Incorporating CPD into the retarding potential yields the linear relation $\frac{1}{\sqrt{V}} = \left(\frac{h}{e} \right) \frac{1}{V} - \frac{\phi_a}{eV}$, in which ϕ_c cancels out entirely. Consequently, the theoretically-intercept is negative ($-\phi_a/e$) and measures the anode work function directly. This indicates something deeper about the Fermi levels of the cathode and anode. The work function itself is the difference of the Vacuum Energy and the Fermi energy. Vacuum Energy is the minimum energy required for an electron to free itself from the surface of the material it is attached to into the surrounding 'vacuum' with no kinetic energy. Fermi Energy is the chemical potential of the electrons inside the material at zero kelvin. When not connected the cathode and anode may have different Fermi levels depending on their material. When they are connected in a circuit, like in our detector, the electron transfer will force their Fermi levels to be equal to keep their charge equal. Key to our experiment, when we apply a voltage V , a singular Fermi level remains but it sits between eV and eV' where $V' = V - \text{CPD}$.

6.3 What Could be Improved

As previously mentioned, the UV light leaked by the filters likely impacted our results. We did not account for the uncertainty when taking the measurements which is directly why our filtered standard deviations are tiny in comparison to the unfiltered run. To account for this, we could have used our unfiltered run as a basis and used the ideal model in figure 12 to calculate what our expected value for each filter should be to help subtract the current caused by the UV light. The uncertainty of the first run could be propagated through this calculation.

7 Conclusions

Ultimately we find our data agrees with the expected values. It is clear that the underestimation of our value (6.58 compared to 6.62) is expected when we cannot cleanly see our reverse current. Our large uncertainty for h is explained easily by how we interpreted the data set and the natural sensitivity of the detector. But more importantly, we found that the work function that in theory should be the photocathode's work function is actually that of the collector anode. This value's deviation is also explainable. The uncertainty's lower bound would tell us that the value of the measured work function is negative. This is entirely not physical but when the experimental reality is understood we realize that the current in our detector does not flow perfectly from the cathode to the anode.

8 Appendix A: Python Script

8.1 Main

```
1 import pandas as pd
2 import numpy as np
3 import matplotlib.pyplot as plt
4 from dataFunctions import decayThreshold, linearFit, fit_bootstrap,
   fit_bootstrap_full
5
6
7 # load the excel sheet
8 excel_file = pd.ExcelFile("Cleaned_PhotoelectricData.xlsx")
9
10 # initialize the array of stopping voltages and arrays for later Planck's
   constant calculation
11 V_stopping = []
12 V_stopping_variation = []
13 h_x = []
14 h_y = []
15
16
17 #Create a function "recipe" for the bootstrap algorithm to use
18 line_recipe = lambda x, p: p[0] * x + p[1]
19
20
21
22 # Physical constants
23 c = 2.99792458e8      # m/s
24 e = 1.602176634e-19   # C
25
26 # index of wavelength in nanometers
27 wavelengths = [254.0e-9, 590.0e-9, 546.0e-9, 405.0e-9, 590.0e-9] # given by
   the manual
28
29 # Iterate directly through sheet names using an if-statement
30 for sheet in excel_file.sheet_names:
31     if sheet.startswith("VvC"):
32         df = pd.read_excel(excel_file, sheet_name=sheet, skiprows=1)
33         df.columns = df.columns.str.strip()
34
35         # Extract V and I arrays
36         V = df["Voltage_(V)"].values
37         I = df["Photocurrent_(uA)"].values #uA is microamperes
38
```

```

39     V = np.array([round(float(item), 1) for item in V]) # our
    measurements were consistent only to the tenths place
40     I = np.array([round(float(item), 1) for item in I]) # this ensures
    that floating point error is minimized
41
42     fastDecayRegion, slowDecayRegion = decayThreshold(I, V)
43
44     #V and I values in the fast decay and the slow decay region
45     V_fastDecay = V[fastDecayRegion]
46     I_fastDecay = I[fastDecayRegion]
47
48     V_slowDecay = V[slowDecayRegion]
49     I_slowDecay = I[slowDecayRegion]
50
51     slope_fast, intercept_fast = linearFit(V_fastDecay, I_fastDecay) # f1
52     slope_slow, intercept_slow = linearFit(V_slowDecay, I_slowDecay) # f2
53
54     # we can now estimate the stopping voltage by averaging our data
55     #  $V = -\text{intercept}/\text{slope}$  when  $I = 0$ 
56     V_1 = -intercept_fast / slope_fast
57
58     # now we need a second point that starts from the same point, but
    intersects with the high v region
59     V_2 = V_1 + (intercept_slow - intercept_fast) / (slope_fast -
    slope_slow)
60
61     V_stoppingAverage = (V_1 + V_2) / 2
62
63     V_stopping.append(float(V_stoppingAverage)) # ensures that the output
    data type is float and not np.float64
64
65     # the error must be propagated from here
66     # each of these saves two columns each consisting of 100000 rows
67     # column zero contains slopes, one contains intercepts
68     boot_fast = fit_bootstrap([slope_fast, intercept_fast], V_fastDecay,
    I_fastDecay, line_recipe, yerr_systematic=0.50)
69     boot_slow = fit_bootstrap([slope_slow, intercept_slow], V_slowDecay,
    I_slowDecay, line_recipe, yerr_systematic=0.50)
70
71
72     boot_slopes_fast = boot_fast[:, 0]
73     boot_intercepts_fast = boot_fast[:, 1]
74
75     boot_slopes_slow = boot_slow[:, 0]
76     boot_intercepts_slow = boot_slow[:, 1]
77

```

```

78     #voltage math again
79     V_1_boot = - boot_intercepts_fast / boot_slopes_fast
80     V_2_boot = V_1_boot + (boot_intercepts_slow - boot_intercepts_fast) /
      (boot_slopes_fast - boot_slopes_slow)
81     V_avg_boot = (V_1_boot + V_2_boot) / 2 # will return 100000 V
      averages which allows us to find the deviation
82
83     # Multiplied by 2 for a 95.44% confidence interval
84     V_stopping_variation.append(2 * np.std(V_avg_boot))
85
86     # I also want to graph the line fits for later analysis
87     plt.figure()
88
89     # plot the data, and two lines
90     plt.scatter(V, I, color='blue', label='Smoothed_Data')
91     plt.plot(V, slope_fast*V + intercept_fast, "r--", linewidth = 1.5,
      label=f"Line_1_($I_f$, $V_1$={V_1:.3f} V)")
92     plt.plot(V, (-intercept_fast/V_2)*V + intercept_fast, "b--",
      linewidth=1.5, label=f"Line_2_($I_s$, $V_2$={V_2:.3f} V)")
93
94     # Mark the calculated stopping voltage average
95     plt.axvline(x=V_stoppingAverage, color='red', linestyle=':', label=f'
      V_avg={V_stoppingAverage:.2f} V')
96
97     # Formatting
98     plt.xlabel("Voltage (V)")
99     plt.ylabel("Photocurrent (\u03bcA)")
100    plt.title('Photocurrent vs. Voltage: Linear Extrapolation Fit')
101    plt.grid(True)
102    plt.legend()
103    plt.show()
104
105
106
107
108    wavelengths = np.array(wavelengths)
109    V_stopping = np.array(V_stopping)
110    V_stopping_variation = np.array(V_stopping_variation)
111
112
113    h_x = 1 / wavelengths
114    h_y = V_stopping #Filter data was not called here as there are too few data
      points
115
116    h_slope, h_intercept = linearFit(h_x, h_y)
117

```

```

118 h = (h_slope * e) / c          #calculate Planck's constant
119 workFunction = abs(h_intercept) #calculate work function, absolute value
    because energy of the work function is always positive
120
121 # deviation of h can be calculated with our uncertainty
122 h_params_mean, h_params_err = fit_bootstrap_full([h_slope, h_intercept], h_x,
    h_y, line_recipe, yerr_systematic=V_stopping_variation)
123
124 h_err = (h_params_err[0] * e) / c
125 workFunction_err = h_params_err[1]
126
127 print(f"Stopping_voltages: {np.round(V_stopping, 3)} +/- {np.round(
    V_stopping_variation, 3)}")
128 print(f"h: {h:e} +/- {h_err:e}")
129 print(f"Work_Function: {workFunction:.3f} +/- {workFunction_err:.3f}")
130
131 # --- Graphing h with uncertainties ---
132 plt.figure(figsize=(8, 6))
133
134 # Plot data points with the bootstrap variation as error bars
135 plt.errorbar(h_x, h_y, yerr=V_stopping_variation, fmt='ko', capsize=4, label=
    "V_stopping +/- Variation")
136
137 # Plot the linear fit
138 fit_line = h_slope * h_x + h_intercept
139 plt.plot(h_x, fit_line, 'r--', label=f"Linear_Fit\nSlope: {h_slope:.3e}")
140
141 plt.xlabel("Inverse_Wavelength_1/$\lambda$(1/nm)")
142 plt.ylabel("Stopping_Potential_(V)")
143 plt.title("Determination_of_Planck's_Constant")
144 plt.legend()
145 plt.grid(True, linestyle="--", alpha=0.6)
146
147 plt.show()

```

8.2 Functions

Here I have attached the functions that are called in the main code for readability. To avoid confusion, first there was a filter on the data which is why there are some instances where the photocurrent data is referred to as I_{smooth} . This filter was removed from the code as it distorted results. Similarly in the line fit function there is another unused filter, this was designed to remove outlier points but also overly distorted the raw data. Finally, there are two different bootstrap functions. The bootstrapFull function returns a linefit and propagated uncertainty. This did not allow for uncertainties to be passed through to calculate the voltage math (V_1 , V_2 , $V_{average}$). Because the bootstrap method was only decided upon later in the development of the code, we were unaware that this method handled linear regression. This did not impact the results.

```
1 import numpy as np
2 from scipy.signal import savgol_filter
3 from scipy import optimize
4
5
6 def decayThreshold(I_smooth, V):
7     # this function will define where the high/low voltage regions are
8
9     # np.gradient computes central differences: (I[i+1] - I[i-1]) / (V[i+1] - V[i-1])
10    slope = np.gradient(I_smooth, V) # this is dI/dV
11
12    steepestIndex = np.argmin(slope) # np.argmin finds index where derivative
13    is most negative (steepest drop)
14    maxDecay = slope[steepestIndex]
15
16    threshold = maxDecay/(2) # defines the max slope value which we still
17    consider to be valid low voltage region
18    # I found that 1/2 returns the closest value of h compared to 1/e
19
20    fastDecayRegion = slope <= threshold # creates an array of bools in our
21    low v
22    slowDecayRegion = (slope >= threshold) & (np.arange(len(V)) >
23    steepestIndex) # creates bool array for high v region
24
25    return fastDecayRegion, slowDecayRegion
26
27
28 def linearFit(x, y):
29     # a linear fit (y = mx + b) will allow us to find slopes and y intercepts
30    of three relevant data sets
31    # the first two will be for our decay regions
32    # the last will be for determining h
```

```

29 slope, y_intercept = np.polyfit(x, y, 1)
30
31
32
33 if len(x) >= 30:      # I found that this may accidentally cut off too many
34 points for this experiment but it may be useful for later labs
35     # a residual verification may be done in case any current reading
36 deviates too far
37     # this is accomplished by using a predicted linear function as a
38 reference
39
40     # firstly we need to see how far away each observed point is away from
41 the predicted ones
42     y_predicted = slope * x + y_intercept
43     residuals = np.abs(y - y_predicted) # np.abs returns an absolute
44 value
45
46     # next we must define a threshold similar to 1/e earlier
47     # the following code says that the only valid points are those
48 whithin
49     # 10% of the range of values observed in current, essentially we
50 claim that the
51     # displacment of individual points from predicted should reflect the
52 total spread of our current data
53     validPoints = residuals <= (0.1 * np.ptp(y))
54     slope, y_intercept = np.polyfit(x[validPoints], y[validPoints], 1)
55
56
57 return slope, y_intercept
58
59
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```

to propagate our measurement errors of photocurrent into stopping voltage

I decided to use the same method used in the example lab report but made some modifications to find V stopping uncertainty

Source - <https://stackoverflow.com/a/21844726>

Posted by Pedro M Duarte, modified by community. See post 'Timeline' for change history

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```

def fit_bootstrap(p0, datax, datay, function, yerr_systematic=0.0):
    errfunc = lambda p, x, y: function(x,p) - y

```

```

65     # Fit first time
66     pfit, perr = optimize.leastsq(errfunc, p0, args=(datax, datay),
full_output=0)
67
68
69     # Get the stdev of the residuals
70     residuals = errfunc(pfit, datax, datay)
71     sigma_res = np.std(residuals)
72
73     sigma_err_total = np.sqrt(sigma_res**2 + yerr_systematic**2)
74
75     # 100000 random data sets are generated and fitted
76     ps = []
77     for i in range(100000):
78
79         randomDelta = np.random.normal(0., sigma_err_total, len(datay))
80         randomdataY = datay + randomDelta
81
82         randomfit, randomcov = \
83             optimize.leastsq(errfunc, p0, args=(datax, randomdataY),\
84                             full_output=0)
85
86         ps.append(randomfit)
87
88     ps = np.array(ps)
89
90     return ps
91
92     # the code below will return slope and intercept uncertainty unlike the
93     # code above which I used to do Voltage error calculations
94     def fit_bootstrap_full(p0, datax, datay, function, yerr_systematic=0.0):
95
96         errfunc = lambda p, x, y: function(x,p) - y
97
98         # Fit first time
99         pfit, perr = optimize.leastsq(errfunc, p0, args=(datax, datay),
full_output=0)
100
101
102         # Get the stdev of the residuals
103         residuals = errfunc(pfit, datax, datay)
104         sigma_res = np.std(residuals)
105
106         sigma_err_total = np.sqrt(sigma_res**2 + yerr_systematic**2)
107
108         # 100000 random data sets are generated and fitted

```



```

109 ps = []
110 for i in range(100000):
111
112     randomDelta = np.random.normal(0., sigma_err_total, len(datay))
113     randomdataY = datay + randomDelta
114
115     randomfit, randomcov = \
116         optimize.leastsq(errfunc, p0, args=(datax, randomdataY),\
117                                full_output=0)
118
119     ps.append(randomfit)
120
121 ps = np.array(ps)
122 mean_pfit = np.mean(ps,0)
123
124 # You can choose the confidence interval that you want for your
125 # parameter estimates:
126 Nsigma = 2. # 1sigma gets approximately the same as methods above
127             # 1sigma corresponds to 68.3% confidence interval
128             # 2sigma corresponds to 95.44% confidence interval
129 err_pfit = Nsigma * np.std(ps,0)
130
131 pfit_bootstrap = mean_pfit
132 perr_bootstrap = err_pfit
133 return pfit_bootstrap, perr_bootstrap

```

9 Appendix B: Raw Data

Table 1: Distance vs. Photocurrent Red Filter, No Retarding Voltage

Distance (cm)	Photocurrent (μA) $\pm 0.5 \mu\text{A}$
10	171.4
11	165.0
12	160.8
13	151.7
14	134.7
15	133.0
16	125.0
17	114.5
18	96.8
19	84.3
20	73.3
21	68.3
22	62.5
23	56.8
24	54.2
25	52.6
26	48.1
27	46.8
28	45.0
29	42.8
30	40.5

The table above displays the relationship between the distance from the light source and the Photoelectric box and the resulting photocurrent that is generated. In this case, a red filter was placed over the lens of the Photoelectric box, and no retarding voltage was added.

Table 2: Voltage vs. Photocurrent No Filter, 25 cm Distance

Voltage (V)	Photocurrent (μA)	Normalized Photocurrent (μA) $\pm 0.5 \mu\text{A}$
0.0	4.55	4.97
0.2	4.55	4.97
0.4	4.55	4.97
0.6	4.55	4.97
0.8	4.54	4.96
1.0	4.54	4.96
1.2	4.54	4.96
1.4	4.53	4.95
1.6	3.59	4.01
1.8	2.27	2.69
2.0	1.28	1.70
2.2	0.58	1.00
2.4	0.12	0.54
2.6	-0.17	0.25
2.8	-0.31	0.11
3.0	-0.36	0.06
3.2	-0.39	0.03
3.4	-0.40	0.02
3.6	-0.41	0.01
3.8	-0.42	0.00
4.0	-0.42	0.00

The table above displays the relationship between the applied voltage and the resulting photocurrent at a distance of 25 cm from the light source. In this case, no filter was placed over the lens of the Photoelectric box.

Table 3: Red Filter: Voltage vs. Photocurrent at 25 cm

Voltage (V)	Photocurrent (μA) $\pm 0.5 \mu\text{A}$
0.0	73.8
0.2	46.0
0.4	26.2
0.6	12.6
0.8	5.2
1.0	1.7
1.2	0.5
1.4	0.2
1.6	0.05
1.8	0.025
2.0	0.0
2.2	0.0
2.4	0.0
2.6	0.0
2.8	0.0
3.0	0.0
3.2	0.0
3.4	0.0
3.6	0.0
3.8	0.0
4.0	0.0

The table above displays the relationship between the applied voltage and the resulting photocurrent at a distance of 25 cm from the light source. In this case, a red filter was placed over the lens of the Photoelectric box.

Table 4: Green Filter: Voltage vs. Photocurrent at 25 cm

Voltage (V)	Photocurrent (μA) $\pm 0.5 \mu\text{A}$	Normalized Photocurrent (μA)
0.0	32.1	31.6
0.2	4.8	4.3
0.4	0.8	0.3
0.6	0.5	0.0
0.8	0.5	0.0
1.0	0.5	0.0
1.2	0.5	0.0
1.4	0.5	0.0
1.6	0.5	0.0
1.8	0.5	0.0
2.0	0.5	0.0
2.2	0.5	0.0
2.4	0.5	0.0
2.6	0.5	0.0
2.8	0.5	0.0
3.0	0.5	0.0
3.2	0.5	0.0
3.4	0.5	0.0
3.6	0.5	0.0
3.8	0.5	0.0
4.0	0.5	0.0

The table above displays the relationship between the applied voltage and the resulting photocurrent at a distance of 25 cm from the light source. In this case, a green filter was placed over the lens of the Photoelectric box.

Table 5: Blue Filter: Voltage vs. Photocurrent at 50 cm

Voltage (V)	Photocurrent (μA) $\pm 0.5 \mu\text{A}$
0.0	142.0
0.2	87.0
0.4	39.5
0.6	14.3
0.8	4.2
1.0	1.2
1.2	0.3
1.4	0.1
1.6	0.0
1.8	0.0
2.0	0.0
2.2	0.0
2.4	0.0
2.6	0.0
2.8	0.0
3.0	0.0
3.2	0.0
3.4	0.0
3.6	0.0
3.8	0.0
4.0	0.0

The table above displays the relationship between the applied voltage and the resulting photocurrent at a distance of 50 cm from the light source. In this case, a blue filter was placed over the lens of the Photoelectric box, and the photocurrent values were not normalized.

Table 6: Red Filter: Voltage vs. Photocurrent at 50 cm

Voltage (V)	Unnormalized Photocurrent (μA) $\pm 0.5 \mu\text{A}$	Normalized Photocurrent (μA)
0.0	16.5	15.8
0.2	10.4	9.7
0.4	6.0	5.3
0.6	3.2	2.5
0.8	1.7	1.0
1.0	1.0	0.3
1.2	0.7	0.0
1.4	0.7	0.0
1.6	0.7	0.0
1.8	0.7	0.0
2.0	0.7	0.0
2.2	0.7	0.0
2.4	0.7	0.0
2.6	0.7	0.0
2.8	0.7	0.0
3.0	0.7	0.0
3.2	0.7	0.0
3.4	0.7	0.0
3.6	0.7	0.0
3.8	0.7	0.0
4.0	0.7	0.0

The table above displays the relationship between the applied voltage and the resulting photocurrent at a distance of 50 cm from the light source. In this case, a red filter was placed over the lens of the Photoelectric Box. Both the unnormalized and normalized photocurrent values are provided, with the normalized values accounting for the background current.

10 References

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